Machine Learning Lab Manual (Notebook Style)

Experiment 2: Logistic Regression on Titanic Dataset

Aim / Objective

To apply logistic regression on the Titanic dataset and predict passenger survival based on various features.

Theory / Introduction

Logistic Regression is a supervised learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values, logistic regression predicts the probability that a given input point belongs to a particular category.

In this experiment, we will use logistic regression to predict whether a passenger survived the Titanic disaster based on features like age, sex, and class. Logistic regression outputs probabilities using the sigmoid function and classifies inputs based on a threshold (usually 0.5).

Software/Tools Required

Python, Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib

Dataset Used

Titanic Dataset (available from [Kaggle Titanic Dataset](https://www.kaggle.com/competitions/titanic/data))

This dataset provides information on the fate of passengers on the Titanic, summarized according to economic status (class), sex, age, and survival.

Key columns:
- `Survived`: 0 = No, 1 = Yes (target variable)
- `Pclass`: Passenger class (1 = upper, 2 = middle, 3 = lower)
- `Sex`: Gender of the passenger
- `Age`: Age in years
- `SibSp`: Number of siblings/spouses aboard
- `Fare`: Ticket fare

Procedure / Steps

1. Load the Titanic dataset.
2. Preprocess the data: handle missing values and encode categorical variables.
3. Split the data into training and test sets.
4. Train a logistic regression model using scikit-learn.
5. Predict the survival of passengers.
6. Evaluate model performance using accuracy and confusion matrix.

7. Visualize prediction results.

Python Step-by-Step Code with Explanations

```
# Step 1: Import required libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Step 2: Load dataset
data =
pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic
.csv")

# Step 3: Data preprocessing
data = data[['Survived', 'Pclass', 'Sex', 'Age', 'Fare']]
data.dropna(inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})  # Encode categorical variable

# Step 4: Feature and target selection
X = data[['Pclass', 'Sex', 'Age', 'Fare']]
y = data['Survived']

# Step 5: Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Step 6: Train Logistic Regression model
model = LogisticRegression()
model.fit(X_train, y_train)

# Step 7: Make predictions and evaluate
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Step 8: Confusion matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
```

```
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

Result / Output

The Logistic Regression model outputs the predicted survival status and the accuracy score. A confusion matrix heatmap is displayed to visualize correct and incorrect predictions.

Conclusion

Logistic Regression successfully predicted survival outcomes on the Titanic dataset. The model demonstrated how categorical and numerical features can be used to build a predictive classifier.

Viva Questions (2–3 per experiment)

1. What does logistic regression predict?
2. Why is logistic regression suitable for binary classification?
3. What is the role of the sigmoid function in logistic regression?